Day 05: Frequency Distribution
Organizing and Understanding Data

100 Days of Data Science

Why We Need Frequency Distribution

Real-world datasets are usually large and messy. Looking at raw data makes it difficult to identify patterns or trends.

Frequency distribution helps by:

- Organizing data in a structured form
- Showing how often values occur
- Making large datasets easy to understand

It is the first step in Exploratory Data Analysis (EDA).

What is Frequency?

Frequency refers to the number of times a particular value appears in a dataset.

Example dataset:

$$2, 3, 3, 4, 5, 5, 5, 6$$

Frequency of $5 = 3$

Frequency Distribution Table

A frequency distribution table lists values along with their frequencies.

Value	Frequency
2	1
3	2
4	1
5	3
6	1

Why This Table is Useful

- Quickly identifies most common values
- Shows data concentration
- Prepares data for graphs

Types of Frequency Distribution

Simple Frequency Distribution

Used when:

- Dataset is small
- Values are not continuous

Each distinct value is listed with its count.

Grouped Frequency Distribution

Used when:

- Dataset is large
- Data is continuous

Data is divided into class intervals such as:

$$0 - -10,\ 10 - -20,\ 20 - -30$$

Each interval shows number of observations within that range.

Example of Grouped Distribution

Class Interval	Frequency
0–10	5
10–20	12
20–30	8
30–40	5

How to Choose Class Intervals

Good class intervals should:

- Cover all data values
- Have equal width
- Not be too many or too few

Poor grouping can hide important patterns.

Connection to Graphical Representation

Frequency distributions are used to create:

- Histograms
- Bar charts
- Frequency polygons

These graphs help visualize data patterns.

Role in Data Science

- Understanding data distribution
- Detecting skewness
- Identifying outliers
- Preparing data for ML models

Common Mistakes

- Using very wide class intervals
- Losing important details
- Misinterpreting grouped data

Final Takeaway

Frequency distribution transforms raw data into organized information.

Good Data Science always starts with well-structured data.